

Range of Complex Engineering Problem Solving

Complex Engineering Problems have characteristic P1 and some or all of P2 to P7

Project: Bike Store Management System

Attributes	Bike Store Management System
P1 Depth of knowledge required (K3-K8)	The project requires knowledge of web-based system design and implementation, including frontend development using HTML, CSS, and JavaScript, backend development using PHP, database design and management using MySQL, and supporting automation/reporting logic using Python. It also requires UI/UX understanding, secure customer data handling, product filtering, recommendation, invoicing, and reporting features (K3, K5, K6, K8).
P2 Range of conflicting requirements	The system must balance rich customer and manager features with performance, speed, usability, data security, and maintainability. Adding browsing, filtering, comparison, cart, dashboard, invoice, and reporting features can increase complexity and may affect system response time.
P3 Depth of analysis required	No unique way exists to design the system. Depth of analysis is required to select suitable technologies, database structure, client-server architecture, user roles, security approach, and feature workflow from several alternatives.
P4 Familiarity of issues	Yes. The design involves issues that may be unfamiliar from previous courses, such as integrating frontend, backend, database, user authentication, product recommendation, digital invoice generation, sales reports, and secure customer information management.
P5 Extent of applicable codes	The project must consider software development standards and good practices related to secure login, data validation, database integrity, privacy of customer information, and reliable digital invoice/report generation.
P6 Extent of stakeholder involvement	Several stakeholders are involved, including store owners, customers, sales staff, and system administrators. Their needs influence the system features, dashboard, product management, booking, invoice, and reporting requirements.
P7 Interdependence between subsystems	The project contains interdependent subsystems such as frontend user interface, backend server logic, MySQL database, customer profile, product inventory, cart/recommendation module, sales dashboard, invoice generation, and report management. A change in one part affects the others.

Course Name: Junior Design | Course Code: CSE299 | Section: 12 | Group: 05